

# AI for Education Initiative

Building Local AI Infrastructure for Teachers and Students

Prepared by William Morris | May 2026

## Executive Summary

We propose establishing a dedicated AI education server at NAS Jiaxing — a local, privacy-focused platform that gives teachers and students direct access to cutting-edge AI tools without relying on external cloud services or VPNs.

### The Problem We Are Solving

Teachers want to use AI in classrooms, but face three barriers:

- Access blocked** — Many AI tools (ChatGPT, Claude, Gemini) are blocked on school networks without VPN
- Data privacy** — Uploading student work to foreign cloud services raises compliance concerns
- Cost and complexity** — Premium AI subscriptions are expensive; teachers don't know where to start

## Our Solution

A local AI server running inside the school network, hosting open-source AI models and educational tools that are:

- Always accessible** — No VPN needed; works on school WiFi
- Fully private** — All data stays inside the school; no external cloud uploads
- Free to use** — Open-source software; no subscription fees
- Educational** — Designed specifically for teaching, not general chat

## What Is Already Working (Demo)

We have built a proof-of-concept running on a personal workstation. This demonstrates that the technology works and the concept is viable.

### Current Demo Machine Specs

| Component | Specification                              |
|-----------|--------------------------------------------|
| CPU       | AMD Ryzen 9 7950X3D (16 cores, 32 threads) |

|                  |                                            |
|------------------|--------------------------------------------|
| RAM              | 96GB DDR5                                  |
| GPU              | AMD RX 7900 XTX (24GB VRAM)                |
| Storage          | 2TB NVMe SSD + 4TB HDD                     |
| OS               | Ubuntu 24.04 LTS                           |
| Network          | School LAN — accessible from any classroom |
| Total Build Cost | ~¥20,000 (personal funds)                  |

This machine currently serves educational content and AI tools to the school network. It proves the concept. For a permanent school solution, we recommend a purpose-built server at roughly half the cost.

# Proposed School Server — Two Build Options

For a permanent, school-owned installation, we researched two viable approaches at the same ~¥10,000 budget. Both are significantly more cost-effective than the ¥20,000 demo build.

★ **Recommended: Used Enterprise Server (Option A)**

A professional 2U rackmount server with 32-core AMD EPYC processor. Enterprise-grade reliability with redundant power, remote management, and massive expansion capacity. This is what real data centers use — repurposed for education.

## Option A: Used Enterprise Server (~¥10,800) ★

All components sourced from Taobao used/server refurb market. Tested before shipping. 1-year seller warranty.

| Component        | Specification                                                                                 | Est. Price |
|------------------|-----------------------------------------------------------------------------------------------|------------|
| Chassis          | GIGABYTE G292-Z20 2U Rackmount Server (used)                                                  | ¥4,700     |
| Motherboard      | MZ22-G20 (SP3 socket, 8x DDR4, 10x PCIe x16)                                                  |            |
| CPU              | AMD EPYC 7532 — 32 cores / 64 threads, 2.4GHz, 256MB L3 cache                                 |            |
| RAM              | 256GB DDR4 ECC RDIMM (8x 32GB, used, Samsung/SK Hynix) — expandable to 512GB                  | ¥2,400     |
| GPU              | 2x NVIDIA Tesla A2 (16GB GDDR6 each, 60W, datacenter AI inference) — expandable to 4x or more | ¥4,000     |
| Storage (System) | 1TB NVMe SSD (used enterprise, e.g., Intel P4510 / Samsung PM981)                             | ¥400       |
| Storage (Data)   | 4TB HDD (used enterprise, e.g., HGST / Seagate Exos)                                          | ¥350       |
| Power            | 2200W Dual Redundant PSU (included with chassis)                                              | Included   |
| Cooling          | 4x Server-grade fans with thermal management (included)                                       | Included   |
| Network          | Dual 10GbE SFP+ ports (onboard) + IPMI remote management                                      | Included   |
| Rails / Rack Kit | Sliding rail kit for server cabinet (optional but recommended)                                | ¥200       |
| OS & Software    | Ubuntu 24.04 LTS (free, open-source)                                                          | ¥0         |
| Total            | 32 cores · 256GB RAM · 32GB VRAM (2x Tesla A2) · Rackmount · 10GbE                            | ~¥12,050   |

💡 Budget-Friendly Alternative: 128GB RAM + 1x Tesla A2 (~¥9,800)

Option B: New Desktop Build (~¥10,200)

| Component            | Specification                            | Est. Price |
|----------------------|------------------------------------------|------------|
| CPU                  | AMD Ryzen 7 7700X (8 cores, 16 threads)  | ¥2,000     |
| RAM                  | 64GB DDR5 (expandable to 128GB)          | ¥2,500     |
| GPU                  | AMD RX 7800 XT (16GB VRAM)               | ¥3,500     |
| Storage              | 1TB NVMe SSD + 4TB HDD                   | ¥1,200     |
| Case/PSU/Motherboard | Standard ATX mid-tower, 750W PSU         | ¥1,500     |
| OS & Software        | Ubuntu 24.04 LTS (free, open-source)     | ¥0         |
| Total                | 8 cores · 64GB RAM · 16GB VRAM · Desktop | ~¥10,200   |

Side-by-Side Comparison

| Feature           | Option A: Used Enterprise                                  | Option B: New Desktop             |
|-------------------|------------------------------------------------------------|-----------------------------------|
| CPU Cores         | ● 32 cores / 64 threads                                    | 8 cores / 16 threads              |
| Form Factor       | ● 2U Rackmount (fits server room)                          | Mid-tower desktop (sits on floor) |
| Power Supply      | ● Dual 2200W redundant (hot-swap)                          | Single 750W (no redundancy)       |
| GPU / VRAM        | ● 2x Tesla A2 (32GB total) — datacenter AI inference cards | RX 7800 XT (16GB) — gaming card   |
| GPU Expansion     | ● 10x PCIe slots — can add 2+ more GPUs later              | 3-4 slots, limited by PSU         |
| Drive Bays        | ● 8x hot-swap 2.5" bays + onboard NVMe                     | 2-3 internal M.2 + 2x 3.5" bays   |
| Network           | ● Dual 10GbE SFP+ (onboard)                                | 1GbE only                         |
| Remote Management | ● IPMI/iKVM (reboot/fix remotely)                          | None (need physical access)       |
| Warranty          | 1 year seller warranty (Taobao)                            | ● 3-year component warranties     |
| Power Efficiency  | 200W TDP (enterprise optimized)                            | ● Lower idle power (105W TDP)     |

|                     |                                        |                                  |
|---------------------|----------------------------------------|----------------------------------|
| Noise               | Louder (server fans, in server room)   | ● Quieter (desktop fans)         |
| RAM Type / Capacity | ● 256GB DDR4 ECC (expandable to 512GB) | 64GB DDR5 (expandable to 128GB)  |
| Vendor Lock-in      | None (standard AMD SP3 platform)       | None (standard AMD AM5 platform) |

### Why We Recommend Option A (Used Enterprise Server from Taobao)

For a school environment, the used enterprise server is the better long-term investment:

- **4x more CPU cores** — 32 EPYC cores vs. 8 desktop cores. Handles more simultaneous users and larger AI models.
- **4x more RAM** — 256GB vs. 64GB. Run multiple large AI models at once without swapping. Every teacher and student gets fast responses.
- **2x professional AI GPUs** — Two Tesla A2 cards (32GB total VRAM) vs. one gaming GPU. Tesla cards are designed for AI inference, not gaming. Lower power, higher efficiency, built for 24/7 datacenter operation. Can expand to 4x cards as needs grow.
- **Rackmount form factor** — sits in the school server room, not under a desk. Professional appearance, secure location.
- **Redundant power** — dual 2200W PSUs. If one fails, the other keeps running. Zero downtime for students.
- **10GbE networking** — dual SFP+ ports onboard. Future-proof when the school upgrades network infrastructure.
- **Massive expansion** — 10 PCIe slots and 8 drive bays. Add more GPUs, storage, or network cards later without replacing the whole machine.
- **Remote management** — IPMI/iKVM lets IT reboot, diagnose, or reinstall from anywhere. No need to walk to the server room.
- **Enterprise reliability** — these servers ran 24/7 for years in data centers. A desktop PC isn't designed for that workload.
- **Proven parts** — all sourced from Taobao's used server market. Sellers test before shipping. 1-year warranty standard.

**Trade-offs:** Slightly louder fans (irrelevant in a server room), DDR4 instead of DDR5 (but 256GB of DDR4 beats 64GB of DDR5 for AI workloads), and used components (but enterprise hardware is built to last 10+ years; these servers have plenty of life left). Tesla A2 cards are used datacenter GPUs — not the latest model, but specifically designed for AI inference and far more efficient than consumer gaming cards for this workload.

## What the Server Will Host

### For Teachers — Professional Development

- **AI Lesson Plan Assistant** — Generate differentiated activities, rubrics, and assessments
- **Text Simplifier** — Adapt reading passages for ESL students by level

- **Discussion Starter Generator** — Create engaging prompts for any topic
- **Vocabulary Builder** — Auto-generate word lists with definitions and example sentences
- **Reading Assessment Creator** — Build comprehension questions with answer keys

### For Students — Learning Tools

- **Educational Games** — Grammar, vocabulary, and critical thinking games
- **Creative Project Generator** — AI-assisted story writing and project ideas
- **AI Tutor** — Students can ask questions and get explanations without leaving the school network
- **Listening Practice** — AI-generated listening exercises with transcripts

### For the School — Infrastructure

- **Internal AI Resource Website** — [ai.nasjiaxing.cn](http://ai.nasjiaxing.cn) — a professional portal for all AI tools
- **Content Management** — IT controls what content is available; everything is curated for education
- **No External Dependencies** — Works even if the school's internet connection is slow or filtered

# Benefits to NAS Jiaxing

## For Teachers

- Save hours on lesson planning and material creation
- Differentiate instruction automatically
- Access PD workshop materials anytime
- No need for personal VPNs or subscriptions
- Data stays inside the school — fully compliant

## For Students

- Personalized learning tools available 24/7 on campus
- Interactive games that reinforce curriculum
- AI writing and creativity support
- Safe environment — no ads, no tracking, no external data sharing

## For the School

- Position NAS Jiaxing as a leader in AI education
- One-time hardware cost vs. ongoing subscription fees
- Full control over content and access
- Scalable — add more capabilities as needed

## For IT & Compliance

- All traffic stays on the school LAN
- No student data leaves the building
- Can be placed under school's existing ICP备案
- Centrally managed and monitored

# Implementation Timeline

| Phase                           | Duration  | Actions                                                           |
|---------------------------------|-----------|-------------------------------------------------------------------|
| Phase 1: Approval & Procurement | 2–3 weeks | Principal and IT approve budget; order server hardware            |
| Phase 2: Setup & Configuration  | 1 week    | Install Ubuntu, configure AI software, set up internal website    |
| Phase 3: Content & Testing      | 1–2 weeks | Load educational tools, test with pilot teachers, gather feedback |
| Phase 4: Launch & Training      | 1 week    | Internal launch, PD workshop for staff, onboarding materials      |
| Phase 5: Expansion              | Ongoing   | Add new tools based on teacher requests, student feedback         |

# Compliance & Security

We are committed to full compliance with Chinese regulations:

- **ICP备案** — The site will operate under the school's existing ICP filing as a subdomain (ai.nasjiaxing.cn)
- **Data Sovereignty** — All data stored locally; nothing sent to foreign servers
- **Content Control** — IT and school leadership review and approve all content before go-live
- **No VPN Required** — Everything runs on the school internal network
- **Firewall Friendly** — Standard HTTP on port 80; no special ports or protocols needed

# What We Need From the School

## Three Requests to Move Forward

### 1. Budget Approval (~¥10,000)

For a dedicated school AI server. We can source used enterprise hardware to stay within budget.

### 2. IT Support — Static IP & Subdomain

A reserved static IP on the school LAN for the server, and a DNS entry for ai.nasjiaxing.cn pointing to that IP.

### 3. ICP Subdomain Authorization

Permission to add ai.nasjiaxing.cn under the school's existing ICP备案.

# About the Demo

A live demo is already running on the school network. You can access it from any device connected to the school WiFi at:

http://10.39.26.217/

Accessible only from the school LAN

This demo shows the landing page, educational games, AI tools, and the general look and feel of what the final site would be.

# About Me — William Morris

I am an English teacher at NAS Jiaxing with a background in technology and a personal interest in AI. I built the demo machine in my own time using personal funds to prove that this is feasible and valuable for our school. I am not asking the school to fund my personal project — I am proposing a separate, school-owned server that serves the entire faculty and student body.



I am happy to manage the technical setup, maintain the server, and train teachers on how to use the tools. My goal is to make NAS Jiaxing a model school for AI-integrated education.

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**Questions?** Contact William Morris at [william.morris@nasjiaxing.cn](mailto:william.morris@nasjiaxing.cn)

Full technical details available upon request · Demo accessible on school network